

MongoDB?

MongoDB is a **NoSQL, document-oriented database**. Unlike traditional SQL databases (like MySQL or PostgreSQL), MongoDB stores data in flexible, JSON-like documents.



Basic Concepts

SQL Term	MongoDB Equivalent	Description
Database	Database	Stores collections
Table	Collection	Stores documents (similar to rows)
Row	Document	JSON-like object with fields and values
Column	Field	Key in a document
Primary Key	<code>_id</code>	Unique identifier for each document



Document Example (Like a Row in SQL)

```
{
  "_id": ObjectId("607d1b2f5c1dfb3c2e9b4e5c"),
  "name": "John Doe",
  "age": 30,
  "email": "john@example.com",
  "interests": ["coding", "music"],
  "address": {
    "city": "New York",
    "zip": "10001"
  }
}
```



MongoDB Core Features

- ✓ **Flexible Schema** – You can store different structures in the same collection.
 - ✓ **Scalable** – Easily handles large volumes of data and horizontal scaling.
 - ✓ **High Performance** – Fast reads and writes.
 - ✓ **Indexing** – Optimized searches using indexes.
 - ✓ **Aggregation** – Powerful queries like GROUP BY, SUM, AVG, etc.
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Common Commands (Mongo Shell Style)

- Download the Shell installer

- Go to the MongoDB Shell download page:
<https://www.mongodb.com/try/download/shell>
- Under **Platform**, select **Windows**.
- Choose the **MSI** (recommended) or **ZIP** package and click **Download**.

- **Install (MSI) or unpack (ZIP)**

- **MSI:**

- Double-click the downloaded `.msi` and follow the prompts.
 - By default it installs to:

```
makefile
CopyEdit
C:\Program Files\MongoDB\mongosh\bin
```

- **ZIP:**

- Extract the ZIP somewhere convenient, e.g.:

```
makefile
CopyEdit
C:\tools\mongosh
```

- Inside you'll find `bin\mongosh.exe`.

- **Add mongosh to your PATH**

1. Press **Win + S**, type “**Environment Variables**”, and choose **Edit the system environment variables**.
2. In the **System Properties** dialog, click **Environment Variables....**
3. Under **System variables**, select **Path** → **Edit....**
4. Click **New**, then paste the full folder path to your `mongosh.exe` location. For example:

```
makefile
CopyEdit
C:\Program Files\MongoDB\mongosh\bin
```

or, if you used the ZIP:

```
makefile
CopyEdit
C:\tools\mongosh\bin
```

5. Click **OK** on all dialogs to save.

- **Verify the installation**

- Open a new Command Prompt or PowerShell window (so it picks up the updated PATH).
- Run:

```
powershell
```

```
CopyEdit
mongosh --version
```

- You should see the shell's version printed, e.g.

```
CopyEdit
1.10.2
```

- **Connect to your MongoDB instance**

- For a local server:

```
powershell
CopyEdit
mongosh
```

- For a remote URI:

```
powershell
CopyEdit
mongosh "mongodb+srv://user:pass@cluster0.example.mongodb.net/mydb"
```

Create a DB and Collection:

```
use schoolDB           // Switch/create DB
db.createCollection("students")
```

Insert a Document:

```
db.students.insertOne({
  name: "Alice",
  age: 15,
  grade: "10"
})
```

Find Documents:

```
db.students.find()           // All
db.students.find({ grade: "10" }) // With condition
```

Update Document:

```
db.students.updateOne(
  { name: "Alice" },
  { $set: { age: 16 } }
)
```

Delete Document:

```
db.students.deleteOne({ name: "Alice" })
```

Basic Command

```
// List all databases
show dbs

// Switch (or create) a database
use myDatabase

// List all collections in current db
show collections

// Insert a document
db.users.insertOne({ name: "Alice", age: 30 })

// Query documents
db.users.find({ age: { $gte: 18 } }).pretty()

// Update documents
db.users.updateOne({ name: "Alice" }, { $set: { age: 31 } })

// Delete documents
db.users.deleteMany({ age: { $lt: 18 } })

// Server status & stats
db.serverStatus()
db.stats()
```

✂ Tools

- **MongoDB Compass:** GUI for MongoDB (good for beginners)
- **Mongo Shell / mongosh:** Command-line tool
- **Mongoose:** ODM library for Node.js
- **Atlas:** MongoDB's cloud service (free tier available)

